

An Open, Validated Finite-Element Drift-Diffusion and Schrödinger–Poisson Framework for ISFET/BioFET Biosensor Simulation with a Self-Consistent Electrolyte Double-Layer Model

A. Author, *Member, IEEE*, B. Author, and C. Author, *Senior Member, IEEE*

Abstract—Ion- and biologically-sensitive field-effect transistors (ISFET/BioFET) are the dominant electronic transducers for label-free, point-of-care disease detection, yet their quantitative design still relies almost entirely on proprietary technology computer-aided design (TCAD) suites that couple awkwardly, if at all, to the electrochemistry of the sensing interface. We present an open, self-contained simulation framework that couples a two-dimensional, steady-state, finite-element drift-diffusion device solver — with an optional self-consistent Schrödinger–Poisson quantum-confinement correction — to a mean-field electrolyte model comprising an amphoteric two-pK site-binding surface and a Gouy–Chapman–Stern double layer. The device solver is first validated against analytic semiconductor physics: the built-in potential matches $V_T \ln(N_A N_D / n_i^2)$ to within 6 mV across three doping decades, the law of mass action holds to 3.3×10^{-16} , the forward-bias ideality factor is 1.005, and terminal currents are conserved to 5×10^{-5} . We then show that the bare device predicts an ideal, unity analyte-to-threshold coupling, and that adding the electrolyte model reproduces the two hallmark non-idealities of real biosensors: a *sub-Nernstian* pH response (31 mV/pH to 57 mV/pH depending on the gate oxide) and the collapse of biomolecule detection once the Debye length falls below the binding distance (a $\sim 400\times$ signal reduction from 1 mM to physiological ionic strength). The framework is intended as a reproducible, license-free platform for biosensor device–interface co-design and for teaching; all code and data are openly released.

Index Terms—ISFET, BioFET, biosensor, drift-diffusion, finite-element method, TCAD, site-binding model, electrical double layer, Debye screening, Schrödinger–Poisson, open-source software.

I. INTRODUCTION

THE field-effect biosensor converts a molecular recognition event into an electrical signal without optical labels. Since Bergveld’s introduction of the ion-sensitive field-effect transistor (ISFET) in 1970 [1], the concept has matured into a broad family of chemically and biologically sensitive FETs (ChemFET, BioFET, ImmunoFET, genetically-modified FET) [2], [3]. The sensing principle is electrostatic: an analyte binding at, or a pH change of, a functionalised insulator surface alters the electrostatic potential at the insulator/electrolyte interface, and this interfacial potential adds directly to the gate

TABLE I
PRINCIPAL SYMBOLS.

Symbol	Meaning
ψ, ϕ_n, ϕ_p	electrostatic / electron / hole quasi-Fermi potential
u, v, w	normalised ψ/V_T , ϕ_n/V_T , ϕ_p/V_T
n, p, n_i	electron, hole, intrinsic carrier density
$V_T = k_B T / q$	thermal voltage
N_A, N_D	acceptor, donor doping
$\Phi_m, \Phi_{\text{ref}}$	metal, reference work function
V_{th}, S	threshold voltage, sub-threshold swing
ψ_0, ψ_d	surface, diffuse-plane potential
σ_0, σ_d	site-binding, diffuse-layer charge density
N_s, K_a, K_b	site density, acid/base dissociation constants
$C_{\text{St}}, C_{\text{dl}}$	Stern, double-layer capacitance
α	van Hal pH-sensitivity parameter
λ_D, I	Debye length, ionic strength
σ_{bio}, d	bound-analyte charge density, distance

bias, shifting the transistor threshold voltage and hence the drain current [4], [5]. Because the readout is a purely electrical, label-free threshold shift compatible with CMOS integration, field-effect biosensors underpin an expanding range of point-of-care diagnostics — pH and metabolic panels, DNA-FET pathogen detection, ImmunoFET cardiac and inflammatory-marker assays, and enzymatic glucose sensing — and have been scaled to massively parallel formats such as semiconductor DNA sequencing [6], [7], [8].

The clinical pull is strong. Cardiac troponin and C-reactive protein for acute-care triage, prostate-specific antigen for oncology, nucleic-acid targets for infectious-disease diagnosis, and glucose for chronic-disease management all demand assays that are fast, quantitative, low-cost, and deployable outside a central laboratory [7], [9], [8]. Field-effect biosensors address these requirements because they are label-free, electrically read, and manufacturable in standard CMOS, and they have been demonstrated for antigen, nucleic-acid, and small-molecule targets [10], [11], [6]. Translating a laboratory demonstration into a manufacturable device, however, requires quantitative modelling of how a given surface chemistry, buffer condition, and device geometry combine to set the sensitivity and detection limit.

The predictive design of such devices requires solving the coupled electrostatics and carrier transport of the semiconductor *together with* the electrochemistry of the electrolyte interface. In practice these two halves are handled

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A. Author, B. Author, and C. Author are with the Department of Electrical Engineering (e-mail: author@institution.edu).

by different communities and different tools. Semiconductor transport is the province of technology computer-aided design (TCAD): commercial drift-diffusion solvers such as Synopsys Sentaurus [12] and the COMSOL Semiconductor Module [13], and, more recently, open-source solvers such as DEVSIM [14] and Sandia’s Charon [15]. The electrochemistry of the oxide/electrolyte interface, meanwhile, is described by site-binding and double-layer theory developed in surface and colloid science [16], [17], [18]. Coupling the two typically requires bespoke scripting on top of a proprietary, license-limited TCAD core, which raises a barrier to reproducibility and to teaching, and makes it difficult for a non-specialist to explore biosensor design trade-offs.

This paper presents a compact, fully open framework that closes that gap. Its contributions are:

- 1) an open, self-contained, finite-element (FEM) two-dimensional drift-diffusion device solver in the quasi-Fermi-potential formulation, solved by a fully coupled Newton method with the exact analytic Jacobian, and supporting heterojunctions, generation–recombination, an insulated work-function-referenced gate, and an optional self-consistent Schrödinger–Poisson quantum-confinement correction (Section III);
- 2) a self-consistent electrolyte-interface model — an amphoteric two- pK site-binding surface coupled to a Gouy–Chapman–Stern double layer — that supplies the sensing physics the device solver lacks (Section IV);
- 3) a quantitative validation of the device solver against analytic semiconductor physics (Section VI); and
- 4) a set of reproducible ISFET/BioFET case studies that recover the characteristic sub-Nernstian pH response and the Debye-screening detection limit, and that map directly onto disease-detection applications (Sections VII–VIII).

We emphasise at the outset that the numerical methods used here are standard; the contribution is not a new algorithm but an *integrated, validated, and openly reproducible* device–electrolyte modelling capability, together with the insight its case studies provide.

II. RELATED WORK

A. From the ISFET to the BioFET

The ISFET removes the metal gate of a MOSFET and exposes the gate insulator to an electrolyte contacted through a reference electrode [1]. Early work established that the device threshold is governed by the interfacial potential set by the insulator surface chemistry, and that different insulators (SiO_2 , Si_3N_4 , Al_2O_3 , Ta_2O_5) yield markedly different pH sensitivities [19], [2]. Extending the ISFET to biomolecular recognition—by immobilising receptors (antibodies, single-stranded DNA, enzymes) on the gate—produces the BioFET, in which target capture changes the interfacial charge [3], [4], [11]. Reviews by Schöning and Poghossian [3], Kaisti [5], and others [20], [9], [8] survey the device variants and their clinical use.

B. Clinical demonstrations

The BioFET has been demonstrated across the diagnostic spectrum. Genetically-modified FETs detect single-nucleotide polymorphisms and DNA hybridisation potentiometrically [11]; the same principle, massively parallelised, underlies ion-based semiconductor sequencing [6]. Label-free immunodetection of protein biomarkers with CMOS-compatible nanowires was shown by Stern *et al.* [10], and subsequent reviews catalogue FET assays for cardiac, oncological, and infectious-disease markers [7], [9], [8]. A recurring theme in these reports is that performance in real (high-ionic-strength) samples is limited less by the transistor than by the electrostatics of the sensing interface—precisely the coupling this paper models.

C. Nanoscale and nanowire biosensors

The drive toward single-molecule sensitivity motivated the silicon-nanowire FET, in which a thin, fully-depleted body maximises the surface-to-volume ratio and the electrostatic response to surface charge [21], [22]. Nair and Alam gave the definitive device-physics analysis of nanowire biosensor settling time and sensitivity [23], and CMOS-compatible nanowire immunosensors were demonstrated by Stern *et al.* [10]. In such thin bodies quantum confinement begins to influence the threshold and the inversion-charge distribution, linking biosensor modelling to the Schrödinger–Poisson methods developed for inversion layers [24], [25], [26].

D. The Debye-screening limit

A defining limitation of electrolyte-gated biosensors is Debye screening: a charged target captured beyond the electrolyte screening length is electrostatically invisible to the channel. Stern *et al.* [27] showed that nanowire-FET response depends strongly on the ratio of the Debye length to the receptor–target distance, and later work developed strategies (low ionic strength, short aptamers, porous membranes) to mitigate it in physiological media [28]. Any predictive BioFET model must therefore include the ionic-strength dependence of the interfacial potential.

E. Site-binding and double-layer theory

The interfacial potential itself is described by the site-binding model of Yates, Levine and Healy [16], generalised to the electrolyte-insulator-semiconductor FET by Fung, Cheung and Ko [17] and cast in an ISFET-friendly form by van Hal, Eijkel and Bergveld [18], whose sensitivity parameter α compactly captures why real devices are sub-Nernstian. The surface charge is balanced by the diffuse double layer described by Gouy–Chapman–Stern theory [29], [30]. These models are mean-field and well-established, and are the natural interface layer to couple to a device solver.

F. Device simulation tools

Numerical drift-diffusion simulation dates to Gummel’s decoupled iteration [31], the Scharfetter–Gummel current discretisation [32], De Mari’s coupled p-n solution [33], and

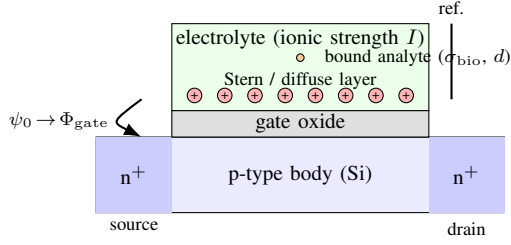


Fig. 1. Modelled ISFET/BioFET stack. The device core solves the semiconductor and oxide; the interface model solves the electrolyte double layer and the bound analyte; they couple at the oxide surface through the interfacial potential ψ_0 , mapped to the gate by Eq. (5).

Selberherr’s canonical treatment [34]. Modern commercial TCAD [12], [13] and open solvers [14], [15] implement these methods at scale. Coupling TCAD to the electrolyte models above, however, is generally done by user scripting rather than as an integrated, openly available capability. Our framework provides that integration in a lightweight, dependency-minimal Python package, with the FEM device core, the electrolyte model, and the coupling in one place; the numerics build on standard components (linear-triangle FEM, coupled Newton, Gauss quadrature, GMSH meshing [35], NUMPY/SCIPY [36], [37]).

G. Compact models and the modelling gap

At the opposite extreme from full TCAD sit closed-form compact models, which express the ISFET threshold as the MOSFET threshold plus an interfacial term $E_{\text{ref}} - \psi_0 + \chi_{\text{sol}}$ [2], [18]. These are invaluable for circuit-level analysis and for extracting sensitivity parameters from data, but they assume a fixed device operating point and cannot resolve two-dimensional electrostatics, short-channel effects, degenerate source/drain regions, or quantum confinement in a thin body. Conversely, full TCAD resolves all of these but treats the electrolyte only through user-supplied boundary conditions. The framework presented here occupies the middle ground: a numerical device solve that captures the two-dimensional electrostatics and the transfer characteristic, coupled to a physically-based electrolyte model, in an open package small enough to be read and modified in full. This niche—between compact models and industrial TCAD—is, to our knowledge, not filled by an existing openly-available, biosensor-oriented tool.

III. DEVICE PHYSICS AND NUMERICAL FRAMEWORK

Figure 1 shows the modelled ISFET/BioFET stack and the coupling between its two physical domains. The semiconductor (source/body/drain) and gate oxide are solved by the device core; the electrolyte, its double layer, and the bound analyte are solved by the interface model; the two meet at the oxide surface, where the interfacial potential ψ_0 is mapped to the gate through the work-function term of Eq. (5).

A. Governing equations

The solver treats the steady-state semiconductor transport problem in the quasi-Fermi-potential formulation. The primary

nodal unknowns are the normalised electrostatic potential $u = \psi/V_T$ and the electron and hole quasi-Fermi potentials $v = \phi_n/V_T$, $w = \phi_p/V_T$, where $V_T = k_B T/q$ is the thermal voltage. Carrier densities follow from Boltzmann statistics,

$$n = n_i e^{u-v}, \quad p = n_i e^{w-u}, \quad (1)$$

so that $np = n_i^2$ at equilibrium ($v = w = 0$). With net doping $C = (N_D - N_A)/n_i$, the coupled system is Poisson’s equation and the two continuity equations,

$$\nabla \cdot (\epsilon \nabla \psi) = -q(p - n + N_D - N_A), \quad (2)$$

$$\nabla \cdot \mathbf{J}_n = +qR, \quad \mathbf{J}_n = q\mu_n n \nabla \phi_n, \quad (3)$$

$$\nabla \cdot \mathbf{J}_p = -qR, \quad \mathbf{J}_p = q\mu_p p \nabla \phi_p, \quad (4)$$

with R the net recombination rate (Shockley–Read–Hall and Auger; zero in the ideal-diffusion limit). All quantities use the De Mari intrinsic scaling [33] so every unknown is $\mathcal{O}(1)$.

B. Finite-element discretisation and coupled Newton solve

The weak forms of (2)–(4) are discretised with linear triangular (P_1) elements and three-point Gauss quadrature, giving a global $3N \times 3N$ residual $\mathbf{F}(u, v, w) = \mathbf{0}$. Because the carrier densities (1) are exponential in the unknowns, the exact analytic Jacobian has a compact block structure whose entries are integrals of the shape functions weighted by e^{u-v} and e^{w-u} ; DDM.SPC assembles this Jacobian analytically and solves the fully coupled system by damped Newton iteration with a per-step increment cap that prevents the exponential terms from overflowing on a cold start. A robust thermal-equilibrium nonlinear-Poisson solve provides the initial guess, and each bias point warm-starts from the previous solution. The terminal current at a contact is extracted as the sum of the natural continuity residuals over the contact nodes, which is conservative by construction.

C. Insulated gate and the work-function transduction principle

An insulated (MOS) gate imposes a Dirichlet condition on the electrostatic potential only; the oxide carries no free carriers, and the quasi-Fermi unknowns of oxide-only nodes are pinned. The gate node potential is referenced to the metal work function Φ_m ,

$$u_{\text{gate}} = \frac{V_{\text{gate}}}{V_T} + \frac{\Phi_{\text{ref}} - \Phi_m}{V_T}, \quad \Phi_{\text{ref}} = \chi + \frac{1}{2}E_g + \frac{1}{2}V_T \ln \frac{N_c}{N_v}. \quad (5)$$

Equation (5) is the hinge of every biosensor model here: a change $\Delta\Phi$ in the effective gate work function shifts the threshold voltage by $\Delta V_{\text{th}} = \Delta\Phi/q$ to first order. In an ISFET the electrolyte supplies that shift through the interfacial surface potential ψ_0 (Section IV), so we identify $\Delta\Phi_{\text{gate}} = -q\psi_0$.

D. Materials, contacts, and carrier statistics

Each mesh region carries its own material (permittivity, affinity, band gap, effective densities of states, mobilities, effective masses, and recombination parameters), so heterojunctions are handled by a position-dependent permittivity

Algorithm 1 Coupled drift-diffusion solve at one bias point

- 1: **Input:** mesh, doping C , contact biases, tolerance τ
- 2: Solve nonlinear Poisson ($v = w = 0$) $\rightarrow$ initial u ; set $v = w = 0$
- 3: **repeat**
- 4: Assemble residual $\mathbf{F}(u, v, w)$ and analytic Jacobian $\mathbf{J}$
- 5: Apply Dirichlet/contact conditions to $\mathbf{F}, \mathbf{J}$
- 6: Solve $\mathbf{J} \Delta \mathbf{x} = -\mathbf{F}$ (sparse direct)
- 7: $\lambda \leftarrow \min(1, \delta_{\max}/\|\Delta \mathbf{x}\|_{\infty})$ $\triangleright$ damping
- 8: $(u, v, w) \leftarrow (u, v, w) + \lambda \Delta \mathbf{x}$
- 9: **until** $\|\Delta \mathbf{x}\|_{\infty} < \tau$
- 10: **return** (u, v, w) ; terminal current from contact residuals

on the Poisson operator, per-element mobilities, and band-edge offsets entering the density exponents via Anderson's electron-affinity rule. Insulator regions (no free carriers) support MOS structures, and the boundary library provides ohmic, insulated-gate, and rectifying Schottky contacts. For degenerate layers—heavily-doped source/drain, strong inversion, or a two-dimensional electron gas—Fermi–Dirac statistics can be selected, entering as a degeneracy shift of the density exponent refreshed by the same outer self-consistency loop as the quantum correction, and reducing to Boltzmann in the non-degenerate limit. These capabilities are what let the same core serve the homojunction diode used for validation (Section VI) and the MOS-gated ISFET used for the biosensor studies.

E. Numerical methods and complexity

The coupled solve is summarised in Algorithm 1. Two features make it robust. First, the equilibrium initial guess is obtained from a nonlinear Poisson solve with $v = w = 0$, so $n = e^u$, $p = e^{-u}$; this places the potential in the basin of attraction of the biased problem before the continuity equations are switched on. Second, each Newton update is damped by a cap $\delta_{\max}$ on the largest increment, which prevents the exponential density terms from overflowing on a cold start while preserving the asymptotic quadratic convergence once the iterate is close to the solution. The linear system at each iteration is sparse ($\mathcal{O}(N)$ non-zeros for a quasi-uniform 2-D mesh of N nodes) and is solved by a sparse direct factorisation; the dominant cost per bias point is therefore that of a handful of sparse solves. Bias sweeps warm-start from the previous converged point, so the number of Newton iterations per step is typically small (2–5) once past threshold. The exponential dependence of the density on the unknowns means the discrete current is naturally of Scharfetter–Gummel character [32]; here it arises from the quasi-Fermi formulation rather than from an edge-based stabilisation.

F. Schrödinger–Poisson correction

For thin, strongly-confined bodies the solver optionally adds a self-consistent quantum-confinement correction. On slices cut across the confinement direction the single-band effective-mass Schrödinger equation

$$\left[-\frac{\hbar^2}{2m^*} \frac{d^2}{dy^2} + U(y) \right] \psi_i = E_i \psi_i \quad (6)$$

is solved by linear FEM; the thermally-weighted subband density is renormalised to preserve the local charge and enters the drift-diffusion assembly as an additive quantum potential $\Lambda = \ln(n_{\text{QM}}/n_{\text{cl}})$ in the density exponent. An outer Gummel-type loop recomputes Λ from the updated potential until convergence, following standard inversion-layer practice [24], [25], [26], [38]; with $\Lambda = 0$ the classical solver is recovered exactly. This capability underpins the future nanowire-BioFET study discussed in Section IX.

IV. ELECTROLYTE INTERFACE MODEL

The bare device solver treats the gate potential as an ideal control and therefore predicts a unity analyte-to-threshold coupling. Real biosensors are governed by the electrolyte interface, which we model with two coupled, well-established ingredients.

A. Site-binding surface charge

A functionalised oxide presents amphoteric hydroxyl sites that protonate or deprotonate according to the local proton activity [16], [17]. For total site density N_s , acid/base dissociation constants K_a, K_b , and surface proton activity $h_s = h_{\text{bulk}} e^{-q\psi_0/k_B T}$, the net surface charge is

$$\sigma_0(\psi_0) = qN_s \frac{h_s/K_a - K_b/h_s}{1 + h_s/K_a + K_b/h_s}, \quad (7)$$

with point of zero charge $\text{pH}_{\text{pzc}} = \frac{1}{2}(\text{p}K_a + \text{p}K_b)$. The surface buffer capacity $\beta_{\text{int}} = -d\sigma_0/d\text{pH}_s$ (large for high N_s and small $\text{p}K$ spread) controls how Nernstian the sensor is.

B. Gouy–Chapman–Stern double layer

The surface charge is balanced by the diffuse layer through a Stern capacitance C_{St} in series with the Gouy–Chapman diffuse charge [29], [30],

$$\sigma_d(\psi_d) = -\sqrt{8\varepsilon_w\varepsilon_0 k_B T n_0} \sinh\left(\frac{q\psi_d}{2k_B T}\right), \quad \psi_0 - \psi_d = \frac{\sigma_0}{C_{\text{St}}}. \quad (8)$$

Charge neutrality $\sigma_0 = -\sigma_d$ closes the system; the module solves the scalar root $\sigma_0(\psi_0) + \sigma_d(\psi_0 - \sigma_0/C_{\text{St}}) = 0$ self-consistently for ψ_0 at each pH using Brent's method (the site charge is bounded, so ψ_0 remains physical).

C. Sub-Nernstian response and the Debye limit

The Nernst limit is $2.303 k_B T/q \approx 59.5 \text{ mV/pH}$ at 298 K; the actual response is reduced by the van Hal sensitivity parameter $\alpha \in (0, 1]$ [18],

$$\frac{d\psi_0}{d\text{pH}} = -2.303 \frac{k_B T}{q} \alpha, \quad \alpha = \left(1 + \frac{2.303 k_B T C_{\text{dl}}}{q^2 \beta_{\text{int}}}\right)^{-1}. \quad (9)$$

Equation (9) follows from a small-signal charge balance at the surface: a change in surface pH perturbs the site charge through the buffer capacity β_{int} , which must be balanced by the double-layer charge through the differential capacitance C_{dl} . The two capacitances form a divider, and the fraction of the ideal Nernstian potential that appears at the surface is $\alpha = \beta_{\text{int}}/(\beta_{\text{int}} + q^2 C_{\text{dl}}/(2.303 k_B T))$, which rearranges to

Algorithm 2 Coupled device–electrolyte biosensor evaluation

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1: Input: oxide  $(N_s, pK_a, pK_b, C_{St})$ , analyte set
    $\{pH, I, \sigma_{bio}, d\}$ , gate reference  $\Phi_0$ 
2: for all analyte states do
3:   Solve  $\sigma_0(\psi_0) + \sigma_d(\psi_0 - \sigma_0/C_{St}) = 0$  for  $\psi_0 \triangleright$  Brent
4:   Add Debye term  $\Delta\psi_{bio} = (\sigma_{bio}/\varepsilon_w\varepsilon_0\kappa)e^{-\kappa d}$ 
5:    $\Phi_{gate} \leftarrow \Phi_0 - q(\psi_0 + \Delta\psi_{bio})$ 
6:   Compute  $I_D(V_G)$  by Algorithm 1 with  $\Phi_{gate}$ 
7:   Extract  $V_{th}$  (constant-current),  $S$ ,  $I_{on}$ 
8: return  $V_{th}$  vs. analyte; sensitivity  $dV_{th}/d(\text{analyte})$ 

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Eq. (9). A surface with abundant, weakly-spread sites (large β_{int}) approaches the Nernst limit; a poorly-buffered surface such as SiO_2 does not. This is the compact explanation for the oxide ordering measured in Table VII.

A biomolecule of charge-sheet density σ_{bio} captured a distance d from the surface contributes, in the linearised Debye–Hückel limit,

$$\Delta\psi_{bio} = \frac{\sigma_{bio}}{\varepsilon_w\varepsilon_0\kappa} e^{-\kappa d}, \quad \lambda_D = \kappa^{-1} = \sqrt{\frac{\varepsilon_w\varepsilon_0 k_B T}{2q^2 N_A I}}, \quad (10)$$

with $\lambda_D \approx 0.30/\sqrt{I[\text{M}]}$ nm at room temperature. When the ionic strength I rises so that $\lambda_D < d$, the response collapses — the Debye-screening limit of Section II.

D. Coupling to the device solver

The self-consistent surface potential ψ_0 enters the device gate as the effective work-function shift $\Delta\Phi_{gate} = -q\psi_0$ (Eq. (5)), and the coupled model then produces a drain-current transfer curve whose threshold is set by the electrolyte. This one-way coupling (interface potential $\rightarrow$ gate) is appropriate in the sub-threshold and low-current regime relevant to sensing, where the channel draws negligible interfacial current and the double layer is in quasi-equilibrium. The full device–electrolyte evaluation is summarised in Algorithm 2: for each analyte state the electrolyte root (8) is solved once, its potential is mapped to a gate work function, and the device transfer curve is computed by Algorithm 1; the threshold and its analyte sensitivity are then extracted from the curve.

V. SOFTWARE ARCHITECTURE

The framework is organised as a small Python package (device core, materials, mesh I/O, FEM primitives, physics assembly, recombination, quantum correction, boundary conditions, solver, and post-processing) plus a set of application scripts. The electrolyte model is a self-contained module that depends only on NUMPY/SCIPY [36], [37]; the device core adds GMSH [35] for unstructured meshes but also ships a built-in structured mesh generator so that simple studies run with no external mesher. Four application programs form a fidelity ladder from an idealised demonstration to a physically realistic biosensor; they are compared in Table II. This separation—an idealised front-end for intuition, a full device for figures of merit, an electrolyte module for the sensing chemistry, and a

TABLE II
THE FOUR APPLICATION PROGRAMS AND THEIR ROLES.

Program	Role	Key physics / output
MOS-cap front-end	transduction demo	Poisson+gate; ψ_s , ΔV_{th}
Transfer-curve model	full device	+continuity; $I_D(V_G)$, S , sensitivity
Double-layer module	electrolyte interface	site-binding + GCS; ψ_0 , α , λ_D
Coupled driver	realistic biosensor	device $\times$ electrolyte; sub-Nernstian dV_{th}/dpH

TABLE III
VALIDATION OF THE DEVICE SOLVER AGAINST ANALYTIC PHYSICS (2-D SI P-N DIODE).

Check	Analytic reference	Result
Built-in potential	$V_T \ln(N_A N_D / n_i^2)$	$\leq 6 \text{ mV err}$
Mass action	$np = n_i^2$	3.3×10^{-16}
Ideality factor	$n = 1$ (ideal)	1.005
Current conservation	$I_a = -I_c$	5×10^{-5}

coupled driver for the realistic device—keeps each concern independently testable and is what makes the workflow suitable for teaching as well as design.

VI. SOLVER VALIDATION

Because the device solver is a research code rather than an established package, we first establish quantitative agreement with analytic semiconductor physics on a two-dimensional silicon p-n diode. Four canonical checks are used (Fig. 2, Table III):

- 1) **Built-in potential.** The equilibrium potential drop matches the analytic $V_{bi} = V_T \ln(N_A N_D / n_i^2)$ to within 6 mV across $N_A = N_D = 10^{15}\text{--}10^{17} \text{ cm}^{-3}$.
- 2) **Law of mass action.** At equilibrium $\max |np/n_i^2 - 1| = 3.3 \times 10^{-16}$, i.e. machine precision.
- 3) **Ideality factor.** The forward-bias current is exponential with ideality $n = 1.005$ (ideal-diffusion value 1.000).
- 4) **Current conservation.** The two terminal currents are equal and opposite to $\max |I_a + I_c|/|I_a| = 5 \times 10^{-5}$.

On a single workstation core the equilibrium solve for the diode completes in a few seconds, a full three-curve ISFET transfer sweep in under three minutes, and the entire electrolyte analysis in well under a second; the coupled device–electrolyte chain over several pH points runs in a few minutes. These times, modest because the problems are two-dimensional and the meshes are small, are what make the framework practical for interactive design exploration and for classroom use rather than only for batch runs.

The one visible departure, at $N_A = N_D = 10^{18} \text{ cm}^{-3}$, where the numeric V_{bi} exceeds the analytic value by $\sim 48 \text{ mV}$, is expected and instructive: the closed-form $V_T \ln(N_A N_D / n_i^2)$ is the non-degenerate (Boltzmann) approximation, and at 10^{18} cm^{-3} the Fermi level approaches the band edge, where that formula itself breaks down and Fermi–Dirac statistics are required. The benchmark thus confirms correct behaviour in

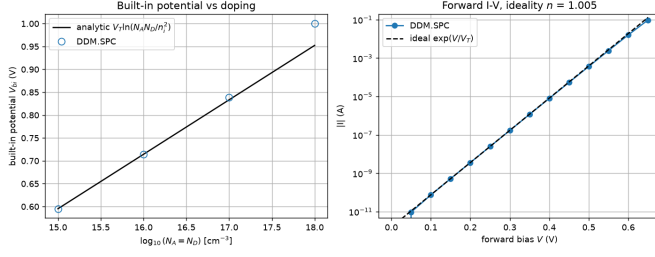


Fig. 2. Validation of the device solver. Left: built-in potential versus doping; the solver (circles) lies on the analytic curve through 10^{17} cm^{-3} . Right: forward-bias I - V on the ideal $\exp(V/V_T)$ law, giving ideality $n = 1.005$.

TABLE IV
SIMULATION PARAMETERS.

Group	Parameter	Value
Device	body doping N_A	10^{17} cm^{-3}
	source/drain N_D	10^{20} cm^{-3}
	drain bias V_D	0.1 V
	width W	1 μm
Electrolyte	temperature T	298 K
	ϵ_w	78.5
	Stern cap. C_{St}	0.2 F/m ² to 0.4 F/m ²
Oxides	Ta ₂ O ₅ (N_s, pK_a, pK_b)	($10^{19}, 2, 4$)
	SiO ₂ (N_s, pK_a, pK_b)	($2 \times 10^{17}, -1, 7$)
Analyte	bound charge σ_{bio}	-0.01 C/m ²
	distance d	3 nm (nominal)

the regime where the analytic law is exact and exhibits the expected onset of its breakdown at degeneracy. Together these checks provide the credibility baseline needed before trusting the device curves that follow.

VII. RESULTS

We now apply the framework to the ISFET/BioFET, progressing from an idealised front-end to the electrolyte-limited full device. The device and interface parameters used throughout are listed in Table IV; the oxide (N_s, pK_a, pK_b) triples are representative literature-consistent values chosen to reproduce the known sensitivity ordering, not fits to a specific fabrication.

A. MOS-capacitor front-end

The transduction principle is isolated in a MOS-capacitor structure (p-Si body under a gate oxide) solved on a structured mesh. Modelling analyte binding as a gate work-function step of 0.20 eV, the silicon surface potential versus gate voltage shifts rigidly by the same amount (Fig. 3): the extracted threshold shift is 200 mV, i.e. near-unity coupling, confirming Eqs. (5)–(7). The characteristic sweeps the silicon surface through accumulation, depletion, and inversion as the gate bias rises, and the analyte step translates the whole curve laterally without changing its shape—the signature of a pure work-function shift and the reason the extracted sensitivity is independent of the chosen readout point in this idealised limit.

B. Full-device transfer characteristics

The complete ISFET (n^+ source/drain, p body, gate oxide) is solved on a tuned short-channel mesh. Figure 4 shows the

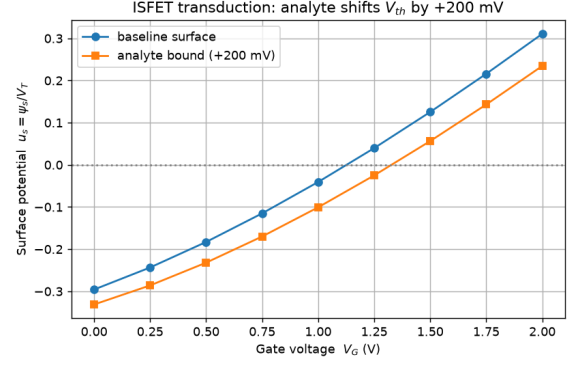


Fig. 3. MOS-capacitor front-end: an analyte-induced gate work-function step rigidly translates the surface-potential characteristic, isolating the ISFET transduction principle.

TABLE V
TRANSFER-CURVE FIGURES OF MERIT ($W = 1 \mu\text{m}$, $V_D = 0.1 \text{ V}$).

$\Delta\Phi$ (eV)	V_{th} (V)	ΔV_{th} (mV)	S (mV/dec)	I_{on} (A)
+0.00	-0.099	0	135	1.15×10^{-3}
+0.10	+0.001	100	135	1.09×10^{-3}
+0.20	+0.101	200	135	1.02×10^{-3}

Sensitivity $dV_{th}/d\Phi = 1.000 \text{ V/V}$ (ideal, unity coupling).

drain-current transfer curves $I_D(V_G)$ for a series of analyte-induced work-function steps, in linear and semilogarithmic form. Table V lists the extracted figures of merit. The analyte-to-threshold coupling is exactly unity ($dV_{th}/d\Phi = 1.000$) — the ideal limit, because the bare device has no electrolyte — and the sub-threshold swing $S \approx 135 \text{ mV/dec}$ exceeds the 60 mV/dec thermal limit, reflecting the short-channel geometry. The semilogarithmic plot (Fig. 4, right) makes the mechanism explicit: below threshold the three curves are parallel and rigidly displaced along the gate axis by the work-function step, while above threshold they converge as the channel enters strong conduction. The elevated swing is a genuine device-physics result of the sub-200 nm channel and the associated drain-induced barrier lowering; a long-channel device would approach the thermal limit, and adding such a geometry is a natural way to strengthen the device-side validation. Because the sensor signal is the sub-threshold *shift* rather than the absolute swing, the transduction remains faithful even when S is degraded, though the resolution analysis of Section VII-D shows a smaller S would improve the detection limit.

C. Electrolyte-limited biosensor

Adding the electrolyte model makes the coupling realistically sub-unity. Figure 5 (left) shows the site-binding surface potential $\psi_0(\text{pH})$ for four representative gate oxides; its slope is the sub-Nernstian sensitivity, summarised in Table VII. The ordering $\text{Ta}_2\text{O}_5 > \text{Al}_2\text{O}_3 > \text{Si}_3\text{N}_4 > \text{SiO}_2$ and the magnitudes (31 mV/pH to 57 mV/pH) reproduce the well-known experimental trend [19], [18]. Figure 5 (right) and Table VIII show the Debye-screening collapse: a fixed bound charge at $d = 3 \text{ nm}$ produces -101 mV at 1 mM but only -0.25 mV at physiological ionic strength ($\lambda_D \approx 0.8 \text{ nm}$),

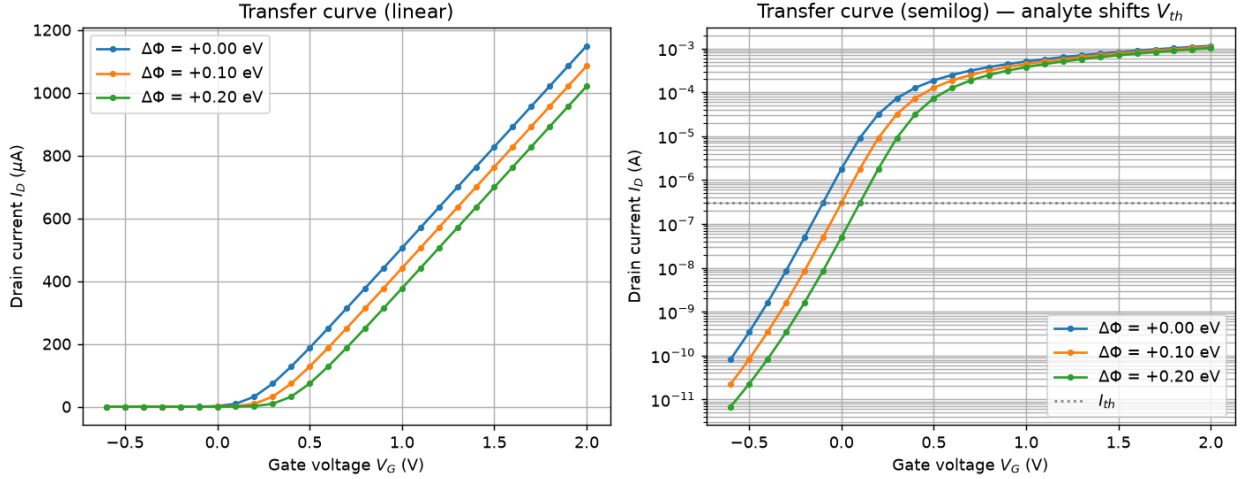


Fig. 4. Full-device ISFET transfer curves $I_D(V_G)$ (linear, left; semilog, right) for analyte-induced work-function steps $\Delta\Phi = 0, 0.1, 0.2$ eV at $V_D = 0.1$ V, $W = 1$ μm . The threshold shift is the sensor signal; the coupling is unity because the bare device lacks the electrolyte.

TABLE VI
FULL DEVICE-ELECTROLYTE CHAIN: ELECTROLYTE-LIMITED
THRESHOLD SENSITIVITY (CONSTANT-CURRENT V_{th} , pH 4/7/10).

Gate oxide	Surface $d\psi_0/d\text{pH}$	Device $dV_{th}/d\text{pH}$
Ta ₂ O ₅	57.4 mV/pH	57.4 mV/pH
SiO ₂	31.6 mV/pH	24.6 mV/pH

TABLE VII
SUB-NERNSTIAN pH SENSITIVITY AT pH 7, $I = 0.15$ M (NERNST LIMIT
59.2 mV/pH).

Gate oxide	Sensitivity (mV/pH)	α
Ta ₂ O ₅	57.4	0.97
Al ₂ O ₃	49.5	0.84
Si ₃ N ₄	46.9	0.79
SiO ₂	31.6	0.53

a $\sim 400\times$ reduction, consistent with the nanowire-FET observations of Stern *et al.* [27]. When the electrolyte model drives the full device solve, the transistor threshold sensitivity is electrolyte-limited (Table VI): 57.4 mV/pH for Ta₂O₅ and 24.6 mV/pH for SiO₂, both below the Nernst limit, unlike the ideal-coupling device of Table V. The near-linear $\psi_0(\text{pH})$ characteristics of Fig. 5 (left) confirm that a single sensitivity slope is a good descriptor over the physiological pH range, and their spread—steep and near-Nernstian for Ta₂O₅, shallow for SiO₂—directly reflects the buffer-capacity ordering through α .

The small SiO₂ offset between the surface slope and the extracted device slope arises from the constant-current threshold definition applied over a wide, three-point pH range together with short-channel non-ideality, and bounds the consistency of the one-way coupling in this regime.

D. Coupling factor and resolution

The two device models bracket the sensitivity. The bare device (Table V) has coupling $g \equiv dV_{th}/d\Phi = 1$, so its pH sensitivity would equal the full Nernst slope; the

TABLE VIII
BIOFET DEBYE SCREENING: BOUND CHARGE -0.01 C/m² AT $d = 3$ nm.

I (M)	λ_D (nm)	Response (mV)
0.001	9.62	−101.3
0.010	3.04	−16.3
0.100	0.96	−0.61
0.150	0.79	−0.25

electrolyte-limited device reduces this to $g \cdot 2.303(k_B T/q)\alpha$ with $\alpha < 1$ (Eq. (9)), giving the measured 24.6 mV/pH to 57.4 mV/pH. The smallest resolvable analyte change follows from the transfer characteristic: a current noise floor $\Delta I/I$ maps to a threshold resolution $\Delta V_{th} \approx S \log_{10}(1 + \Delta I/I)$ through the sub-threshold swing S , and thence to a surface-potential resolution $\Delta\psi_0 = \Delta V_{th}/g$. With the simulated $S \approx 135$ mV/dec and a conservative 1% current resolution, the device resolves $\Delta V_{th} \approx 0.6$ mV, i.e. a pH resolution of ~ 0.01 – 0.02 for a near-Nernstian oxide. This chain—noise floor, sub-threshold swing, coupling, sensitivity—is exactly what the framework makes computable end-to-end, and it identifies S (device geometry) and α (surface chemistry) as the two independent levers on the detection limit.

VIII. DISCUSSION

A. Design implications for disease detection

The results map directly onto label-free diagnostic devices. Because the analyte alters ψ_0 and hence the threshold via Eq. (5), the framework lets a designer trade off the sensing chemistry and the device electrostatics in one place: (i) for *pH* and *metabolic* assays, oxide choice sets the sensitivity through α (Table VII); (ii) for *DNA-FET* and *ImmunoFET* assays, the Debye limit (Table VIII) dictates that low ionic strength or short receptor–target distances are required for the target charge to be visible; (iii) for *enzymatic (glucose)* sensors, the local pH change produced by the enzymatic reaction is read out as in the pH case. The unity-coupling device

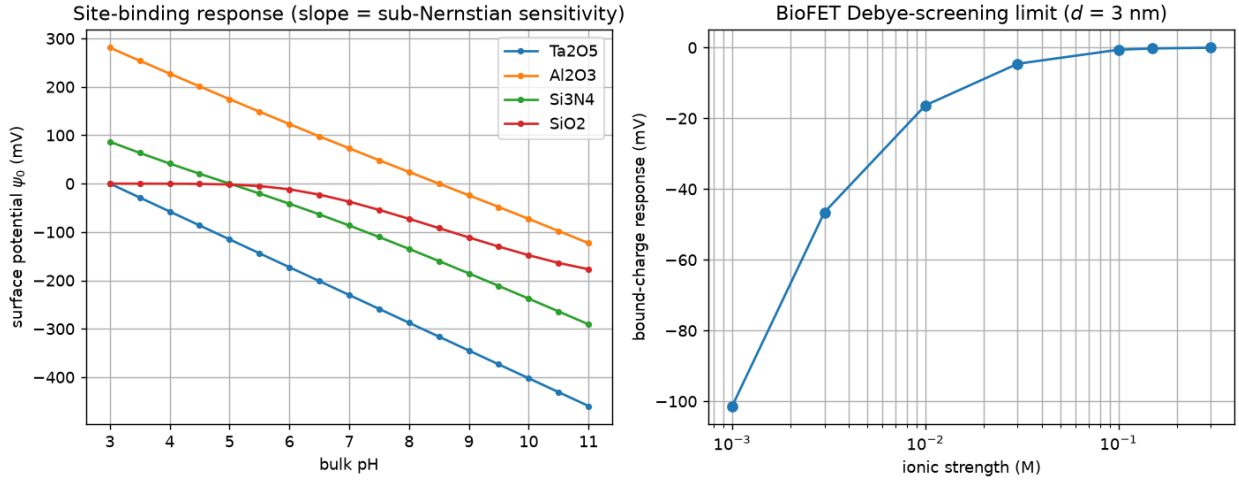


Fig. 5. Electrolyte double-layer model. Left: site-binding surface potential ψ_0 (pH); the slope is the sub-Nernstian sensitivity. Right: the Debye-screening limit — a bound charge at $d = 3$ nm becomes undetectable as increasing ionic strength shrinks λ_D below d .

model (Table V) and the electrolyte-limited model bracket the achievable sensitivity, and the difference between them quantifies exactly how much signal the double layer removes.

B. Worked example: an ImmunoFET in physiological buffer

To illustrate the model as a design aid, consider a cardiac-troponin ImmunoFET operated in physiological buffer ($I = 0.15$ M). The Debye length is $\lambda_D \approx 0.79$ nm (Table VIII). An antibody-captured antigen sits well beyond this length: with a typical antibody footprint the effective charge centroid is $d \approx 5$ nm to 12 nm from the oxide, so the screening factor e^{-d/λ_D} is of order 10^{-3} or smaller, and the model predicts a surface-potential signal below the 0.6 mV resolution estimated in Section VII-D—i.e. the assay is essentially undetectable at physiological ionic strength. Reducing the buffer to $I = 1$ mM raises λ_D to 9.6 nm, bringing $d \lesssim \lambda_D$ and the predicted signal into the 10 mV to 100 mV range (Table VIII). The model thus reproduces, quantitatively, the well-documented requirement to sense in low-ionic-strength media or with charge-proximity strategies [27], [28], and lets a designer estimate the required buffer dilution before fabrication.

C. A biosensor design map

The interplay of binding distance and ionic strength can be condensed into a single design chart. Figure 6 plots the predicted surface-potential response for a fixed bound charge as a function of the analyte binding distance d , for four ionic strengths, against the 0.6 mV resolution floor estimated in Section VII-D. The region above the floor is the detectable design space. At physiological ionic strength it is confined to $d \lesssim 2$ nm—smaller than a typical antibody—whereas at 1 mM the entire 0.5 nm to 12 nm range is detectable. Such a chart, produced in under a second by the electrolyte module, turns the abstract Debye constraint into an actionable specification on receptor size and buffer condition.

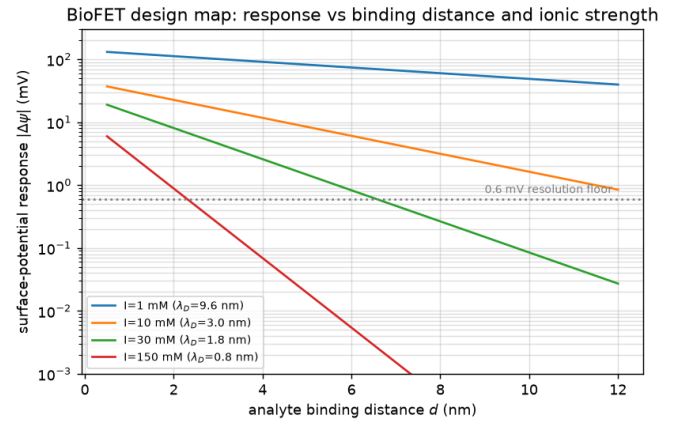


Fig. 6. BioFET design map: predicted surface-potential response versus analyte binding distance d for four ionic strengths, against the 0.6 mV resolution floor (dotted). The detectable design space (above the floor) collapses toward the surface as ionic strength rises.

D. Design guidelines

Table IX distills the results into practical guidance for four common assay classes. The recurring message is that device and interface must be co-designed: a high- α oxide maximises pH sensitivity but does nothing for a charge-based assay that is Debye-limited, while a charge-based assay in physiological buffer demands either a reduced ionic strength, a short receptor–target distance, or a Debye-length-extending strategy such as a porous membrane [28]. The framework makes these trade-offs quantitative in a single tool.

E. Practical non-idealities

Several effects important in real ISFET operation are outside the present steady-state, mean-field scope but are worth situating relative to the model. *Temperature* enters through V_T , n_i , the mobilities, and the electrolyte dissociation constants; the framework can be re-run at any temperature by setting

TABLE IX
DESIGN GUIDANCE DERIVED FROM THE MODEL FOR COMMON ASSAY CLASSES.

Assay		Dominant limit	Design lever
pH metabolic	/	buffer capacity α	high- N_s (Ta ₂ O ₅ /Al ₂ O ₃) oxide
DNA-FET		Debye screening	low I ; short probe; $d < \lambda_D$
ImmunoFET		Debye + coverage	low I ; small receptor; high capture density
Enzymatic (glucose)		local pH gain	high- α oxide + enzyme loading

TABLE X
POSITIONING AGAINST REPRESENTATIVE DEVICE-SIMULATION TOOLS.

Tool	Open	Light	Quantum	Electrolyte
Sentaurus [12]	no	no	yes	scripted
COMSOL [13]	no	no	yes	scripted
DEVSIM [14]	yes	yes	partial	scripted
Charon [15]	yes	no	partial	scripted
This work	yes	yes	yes (S-P)	built-in

the material and interface parameters accordingly, but does not yet model self-heating. *Drift and hysteresis*—the slow, sub-Nernstian wander of the ISFET response attributed to buried-site hydration and ion migration in the insulator [39]—are dynamic phenomena requiring a transient solve and an extended interface model, and set a practical floor on the resolution estimated in Section VII-D. The *reference electrode* is assumed ideal here, contributing a constant offset; in integrated or reference-electrode-free designs its non-idealities must be added to the gate term of Eq. (5). Compact ISFET models developed for circuit simulation capture several of these effects phenomenologically [40], [41], and could be combined with the physical device model presented here. Finally, the enhanced sensitivity reported for aggressively-scaled nanoscale FET biosensors has both electrostatic and quantum origins [42], [23]; disentangling them is precisely the role envisaged for the Schrödinger–Poisson capability of Section III.

F. Relation to existing tools

Compared with commercial TCAD [12], [13], the present framework is far more limited in scope but is open, lightweight, and ships the electrolyte coupling as a first-class component rather than as user scripting. Compared with open device solvers [14], [15], it adds the integrated site-binding/double-layer front-end and a worked biosensor workflow. Table X positions it against representative tools. Its intended role is complementary: a reproducible, license-free platform for biosensor device–interface co-design and for teaching, and a transparent reference against which larger tools can be cross-checked.

IX. LIMITATIONS AND FUTURE WORK

Several limitations bound the present scope and define the roadmap. The device solver is steady-state, so binding

kinetics, impedance/ C - V , and low-frequency noise are out of scope; a transient continuity solve is the natural extension. There is no optical generation term, so photodetector-style biosensors are not yet covered. The electrolyte model is mean-field: the two- pK site-binding parameters (N_s , pK_a , pK_b) are representative values chosen to reproduce the known oxide ordering and sensitivity ranges rather than fitted to a specific fabrication, and the coupling to the device is one-way. The mean-field electrolyte treatment also neglects ion correlations, finite ion size, and Stern-layer dispersion, which become significant at high ionic strength and for multivalent ions; more elaborate modified-Poisson–Boltzmann descriptions could be substituted for Eq. (8) within the same coupling scheme. Most importantly, the biosensor results are validated against the analytic Nernst and Debye limits and the device solver against analytic diode physics, but not yet against measured device data; experimental calibration—extracting (N_s , pK_a , pK_b , C_{St}) from a titration and the device parameters from a fabricated transistor—is required before quantitative, device-specific design claims can be made. Until then the framework is best regarded as a physically-grounded design-space explorer and teaching tool rather than a predictive replacement for calibrated TCAD. Finally, the Schrödinger–Poisson capability of Section III enables a quantum-corrected nanowire-BioFET study — quantifying how carrier confinement in a few-nanometre body modifies sensitivity [23], [24] — which we identify as the most promising direction for a novel, physics-driven result.

X. CONCLUSION

We have presented an open, validated, finite-element drift-diffusion and Schrödinger–Poisson framework for ISFET/BioFET simulation, coupled to a self-consistent site-binding and Gouy–Chapman–Stern electrolyte model. The device solver reproduces analytic semiconductor physics quantitatively (built-in potential to 6 mV, mass action to 10^{-16} , ideality 1.005, current conservation to 10^{-5}), and the coupled model recovers the two defining non-idealities of real biosensors — sub-Nernstian pH response and the Debye-screening detection limit — from first principles. Because the framework is compact, dependency-minimal, and openly released with all scripts and data, it offers a reproducible platform for biosensor device–electrolyte co-design and for teaching, and a foundation for the quantum-corrected nanowire-BioFET study outlined above. More broadly, the work illustrates a modelling pattern that is likely to recur as electronic biosensors proliferate: a validated but lightweight device solver, coupled to a physically-based interface model, can turn qualitative design intuition into quantitative, reproducible predictions without the overhead of a full commercial TCAD flow. We hope the openness of the framework lowers the barrier for other groups to extend it—toward transient operation, additional surface chemistries, and, ultimately, experimental calibration.

REPRODUCIBILITY AND CODE AVAILABILITY

All source code, meshes, material files, and the scripts that generate every figure and table in this paper are openly available in the project repository, along with build instructions.

The four programs used here are the MOS-capacitor front-end, the full-device transfer-curve model, the electrolyte double-layer module, and the coupled device-electrolyte driver; the validation benchmark of Section VI is provided as a standalone script.

APPENDIX A

DE MARI SCALING AND THE ANALYTIC JACOBIAN

The solver works in De Mari intrinsic units [33]: potentials are scaled by V_T , lengths by the intrinsic Debye length $L_D = \sqrt{\epsilon V_T / q n_i}$, and carrier densities by n_i , so that every unknown is $\mathcal{O}(1)$ and the assembled system is well conditioned across many doping decades. In these units the Poisson residual for node i is $\int (\nabla N_i \nabla u + N_i (n - p - C)) d\Omega$ and the two continuity residuals are $\int \mu n \nabla N_i \cdot \nabla v d\Omega$ and the hole analogue.

Because $n = e^{u-v}$ and $p = e^{w-u}$, the element Jacobian has the block form

$$J = \begin{bmatrix} K_{uu} & K_{uv} & K_{uw} \\ K_{vu} & K_{vv} & 0 \\ K_{wu} & 0 & K_{ww} \end{bmatrix}, \quad (11)$$

with, for example, $(K_{uu})_{ik} = \int \nabla N_i \cdot \nabla N_k d\Omega + \int N_i N_k (e^{u-v} + e^{w-u}) d\Omega$ and $(K_{uv})_{ik} = - \int N_i N_k e^{u-v} d\Omega$. All entries are evaluated analytically at the three-point Gauss nodes, so the Newton iteration is exact and retains quadratic convergence. Enabling generation-recombination makes the previously-zero K_{vw}, K_{wv} blocks non-zero through the analytic derivatives $\partial R / \partial n$, $\partial R / \partial p$; the quantum and Fermi-Dirac corrections enter as frozen additive shifts in the density exponents and hence leave the Jacobian structure unchanged within a Newton solve, being refreshed by the outer self-consistency loop.

APPENDIX B

REPRODUCING THE RESULTS

Each figure and table is produced by a single script: the validation benchmark (Fig. 2, Table III); the MOS-capacitor front-end (Fig. 3); the transfer-curve model (Fig. 4, Table V); and the electrolyte/coupled driver (Figs. 5, 6, Tables VII–VI). The device scripts require the device package and GMSH; the electrolyte module requires only NUMPY/SCIPY. Run times are reported in Section VI.

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